

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12389436
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Ikeda; Nobuhiro

Communication apparatus, control method, and computer-readable storage medium

Abstract

A communication apparatus transmits a radio frame complying with an IEEE802.11 standard series to another communication apparatus, or receives the radio frame from the other communication apparatus. Here, a MAC (Media Access Control) frame of the radio frame includes information concerning scheduling of transmission opportunities in which contention is restricted based on a time.

Inventors:	Ikeda; Nobuhiro (Kanagawa, JP)
Applicant:	CANON KABUSHIKI KAISHA (Tokyo, JP)
Family ID:	77672096
Assignee:	CANON KABUSHIKI KAISHA (Tokyo, JP)
Appl. No.:	17/886642
Filed:	August 12, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220386321 A1	Dec. 01, 2022

Foreign Application Priority Data

JP	2020-043334	Mar. 12, 2020
----	-------------	---------------

Related U.S. Application Data

continuation parent-doc WO PCT/JP2021/009591 20210310 PENDING child-doc US 17886642

Publication Classification

Int. Cl.: H04W72/54 (20230101); H04W72/20 (20230101)

U.S. Cl.:

CPC H04W72/54 (20230101); H04W72/20 (20230101);

Field of Classification Search

CPC: H04W (72/54); H04W (72/20); H04W (72/12); H04W (74/02); H04W (74/0816); H04W (74/002); H04W (84/12)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10021646	12/2017	Takahashi et al.	N/A	N/A
10278128	12/2018	Takahashi et al.	N/A	N/A
2018/0332534	12/2017	Hou	N/A	H04W 52/0229
2019/0090220	12/2018	Li et al.	N/A	N/A
2021/0329500	12/2020	Cariou	N/A	H04W 74/002
2022/0346149	12/2021	Ikeda	N/A	N/A
2023/0022414	12/2022	Kim	N/A	H04L 1/1896
2023/0180208	12/2022	Cavalcanti	370/329	H04W 72/0446
2024/0015780	12/2023	Dong	N/A	H04W 74/002

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
105009491	12/2014	CN	N/A
3166257	12/2016	EP	N/A
2018-050133	12/2017	JP	N/A
2018-207308	12/2017	JP	N/A
2020/013874	12/2019	WO	N/A

OTHER PUBLICATIONS

Yang, W. et al., “Survey on IETF 6TiSCH Industrial Internet of Things: Standards, Key Technologies and Platforms” Journal of Frontiers of Computer Science and Technology (Dec. 2019) pp. 361-376, vol. 14, No. 3. cited by applicant

Chinese Office Action issued by the China National Intellectual Property Administration on Aug. 29, 2024 in corresponding CN Patent Application No. 202180019870.1 with English translation. cited by applicant

Nasrallah, A. et al., “Ultra-Low Latency (ULL) Networks: A Comprehensive Survey Covering the IEEE TSN Standard and Related ULL Research” arXiv:1803.07673v1 (Mar. 2018) pp. 1-59, XP080861399. cited by applicant

Cavalcanti, D. et al., “Extending Accurate Time Distribution and Timeliness Capabilities Over the Air to Enable Future Wireless Industrial Automation Systems” Proceedings of the IEEE (Jun. 2019) pp. 1132-1152, vol. 107, No. 6, XP011725970. cited by applicant

Extended European Search Report issued by the European Patent Office on Feb. 14, 2024 in

corresponding EP Patent Application No. 21768937.1. cited by applicant
International Search Report issued by the Japan Patent Office on Jun. 8, 2021 in corresponding International Application No. PCT/JP2021/009591, with English translation. cited by applicant
Cavalcanti, D. et al., "Capabilities to support Time-Aware Scheduling in 802_11be", IEEE 802.22-19/1933r1, IEEE, Internet <URL: <https://mentor.ieee.org/802.11/dcn/19/11-19-1933-01-00be-capabilities-to-support-time-aware-scheduling-in-802-11be.pptx>> (Nov. 2019) pp. 12-17. cited by applicant

Primary Examiner: Sheikh; Ayaz R

Assistant Examiner: Thawng; Mang Boi

Attorney, Agent or Firm: Carter, DeLuca & Farrell LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of International Patent Application No. PCT/JP2021/009591, filed Mar. 10, 2021, which claims the benefit of Japanese Patent Application No. 2020-043334 filed Mar. 12, 2020, both of which are hereby incorporated by reference herein in their entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a communication apparatus, a control method, and a computer-readable storage medium and, more particularly, to an information sharing technique for a scheduling function in wireless communication.

Background Art

(2) The IEEE802.11 standard series is known as the communication standard of a wireless LAN (Local Area Network) defined by IEEE (Institute of Electrical and Electronics Engineers). The IEEE802.11 standard series includes IEEE 802.11a/b/g/n/ac/ax standards, and there has been examined a new standard to improve the peak throughput and frequency use efficiency, as compared with the conventional standards. In, for example, the IEEE802.11ax standard, it is possible to obtain high peak throughput using OFDMA (Orthogonal Frequency Division Multiple Access) (see Japanese Patent Laid-Open No. 2018-050133).

(3) In recent years, to further improve the throughput and the frequency use efficiency, an IEEE802.11be standard has been defined as a new standard. One target for this working is to implement reliable low-latency (RLL) communication.

(4) The present invention provides a communication control technique for implementing such reliable low-latency communication.

SUMMARY OF THE INVENTION

(5) A communication apparatus according to one aspect of the present invention is a communication apparatus comprising a communication unit configured to transmit a radio frame complying with an IEEE802.11 standard series to another communication apparatus, or receive the radio frame from the other communication apparatus, wherein a MAC (Media Access Control) frame of the radio frame includes information concerning scheduling of transmission opportunities in which contention is restricted based on a time.

(6) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments of the invention and, together with the description, serve to explain principles of the invention.
- (2) FIG. 1 is a view showing an example of the configuration of a network according to an embodiment.
- (3) FIG. 2 is a block diagram showing an example of the hardware arrangement of a communication apparatus.
- (4) FIG. 3 is a block diagram showing an example of the functional arrangement of the communication apparatus.
- (5) FIG. 4 is a flowchart illustrating an example of the procedure of processing executed by an AP at the time of STA connection.
- (6) FIG. 5 is a flowchart illustrating an example of the procedure of data reception processing.
- (7) FIG. 6 is a flowchart illustrating an example of the procedure of data transmission processing.
- (8) FIG. 7 is a sequence chart showing an example of the flow of messages transmitted/received between the AP and an STA at the time of DL communication.
- (9) FIG. 8 is a sequence chart showing an example of the flow of messages transmitted/received between the AP and the STA at the time of UL communication.
- (10) FIG. 9 is a view showing an example of the structure of TAS capability information.
- (11) FIG. 10 is a view showing an example of the structure of TAS capability information.
- (12) FIG. 11 is a table for explaining the setting values of the TAS capability information.
- (13) FIG. 12 is a table for explaining the setting values of the TAS capability information.
- (14) FIG. 13 is a flowchart illustrating an example of the procedure of processing when broadcasting the TAS capability information by a Beacon frame.
- (15) FIG. 14 is a view for explaining an example of a primary channel in a 5-GHz band.
- (16) FIG. 15 is a view for explaining TAS traffic processing.
- (17) FIG. 16 is a timing chart for explaining TAS traffic processing.

DESCRIPTION OF THE EMBODIMENTS

(18) Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the claimed invention. Multiple features are described in the embodiments, but limitation is not made to an invention that requires all such features, and multiple such features may be combined as appropriate. Furthermore, in the attached drawings, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.

(19) (Network Configuration)

(20) FIG. 1 shows an example of the configuration of a network according to an embodiment. As parts of the network of FIG. 1, a wireless communication network (BSS **101**) in which an AP **102** provides a wireless communication service and a wireless communication network (BSS **111**) in which an AP **112** provides a wireless communication service are included. Note that the AP is an acronym for Access Point, and the BSS is an acronym for Basic Service Set. A wireless terminal (STA) can be connected to the AP that provides a communicable area including a position at which the self-apparatus exists, thereby joining a BSS provided by the AP. The STA can perform transmission/reception of a radio frame in the joined network. In the example shown in FIG. 1, since an STA **103** exists in both the communicable areas of the APs **102** and **112**, it can be connected to one of the APs to join one of the BSSs **101** and **111**. Note that an environment where the communicable areas of a plurality of APs overlap each other will sometimes be referred to as an OBSS environment hereinafter. That is, in this embodiment, the STA **103** exists in the OBSS

environment. Note that the STA **103** is connected to the AP **102**, and a signal from the AP **112** can be an interference signal for the STA **103**. Furthermore, since an STA **113** exists in the communicable area of the AP **112**, it can be connected to the AP **112** to join the BSS **111**.

(21) The APs **102** and **112** are connected by bridges **104**. The AP and the bridge or the bridge and the other bridge can be connected by, for example, a wired line. However, in at least part of the connection section between the APs **102** and **112**, a wireless line may be used. This arrangement can form one network by merging the wireless networks formed by the APs **102** and **112** and the wired networks formed by the bridges **104** and the like.

(22) A sensor apparatus **107** is a sensor connected to the network via the bridge **104**, and is configured to, for example, output environment data collected by a sensor function to another apparatus. In this embodiment, the sensor apparatus **107** transmits the collected data to the STA **103** at a predetermined cycle. A remote apparatus **108** is a device that is connected to the network via the bridge **104** and operates in accordance with a remote operation instruction from another apparatus. The remote apparatus **108** is an end device such as a industrial robot or a medical operating robot that requires low-latency data communication by an RTA (Real Time Application). This embodiment assumes that the remote apparatus **108** operates in accordance with a remote operation instruction from the STA **103**.

(23) In this network system, for example, it is important that the STA **103** can reliably receive the data from the sensor apparatus **107** at the predetermined cycle, and a remote operation instruction signal from the STA **103** reaches the remote apparatus **108** with low latency. However, in the conventional wireless LAN, since an apparatus for transmitting data confirms that no radio resources are used in the periphery and transmits a signal during a period in which no radio resources are used, the apparatus cannot perform communication while another apparatus uses the radio resources. Therefore, to ensure transmission/reception of a signal at the predetermined cycle and enable communication with low latency, it is necessary to introduce some communication control function. As one solution for this, a Time-Aware Scheduling (TAS) scheme defined by the IEEE802.1Qbv standard for making it possible to appropriately execute transmission/reception of predetermined data such as data having a strict latency requirement can be introduced. The TAS scheme is a time-aware scheduling scheme for causing transmission/reception of a signal at a predetermined time with reference to the time. In this embodiment, in the network, communication adopting the TAS scheme is performed, thereby appropriately executing communication at the predetermined cycle or low-latency communication. To do this, assume that the respective apparatuses such as the AP **102**, the STAs **103** and **113**, and the bridge **104** support the TAS scheme. However, assume that the AP **112** does not support the TAS scheme. Note that the apparatuses other than the AP **112** further support the standard (IEEE802.1AS) synchronized with the same reference clock in the network.

(24) Predetermined data transmitted/received by scheduling using the TAS scheme will be referred to as TAS traffic data hereinafter. In this example, TAS traffic data are packet data transmitted/received in consideration of latency control between the end devices. To control such transmission/reception, the network includes a network management apparatus **105** and a schedule management apparatus **106**. The network management apparatus **105** executes latency control concerning TAS traffic data. The schedule management apparatus **106** performs schedule management concerning the TAS traffic of the end device in the network that uses TAS traffic data. The AP **102** transmits/receives data to/from the network management apparatus **105** via the bridge **104**. Furthermore, the AP **102** exchanges TAS information with the schedule management apparatus **106** that performs schedule management of the whole network.

(25) The network management apparatus **105** collects requirements concerning time-aware scheduling (TAS) from the end devices such as the STA **103**, the sensor apparatus **107**, and the remote apparatus **108**. Based on the requirements, for example, specific apparatuses that transmit/receive the TAS traffic data, a time interval at which the data is transmitted, the magnitude

of allowed latency, and the like are designated. The network management apparatus **105** detects the network topology of the network to be controlled by the self-apparatus. The network topology can be a relationship indicating a specific apparatus to which each apparatus in the network is connected. Note that “connected” indicates a state in which the apparatuses are physically, directly connected or a state in which a logical link is established regardless of a physical connection arrangement. Based on the collected requirements, the network management apparatus **105** calculates, for each of transmission and reception of the TAS traffic data to be executed, an end-to-end path between the end devices. Then, the network management apparatus **105** performs scheduling by determining, based on the calculation result, a timing at which each apparatus transmits a signal, thereby notifying the schedule management apparatus **106** of the scheduling result.

(26) The schedule management apparatus **106** transmits a message including the TAS information to the bridge **104** and the AP **102**, thereby setting a transmission/reception schedule. The schedule management apparatus **106** also notifies the end devices (the STA **103**, the sensor apparatus **107**, and the remote apparatus **108**) of the TAS information. At this time, each of the end devices (the STA **103**, the sensor apparatus **107**, and the remote apparatus **108**) operates as a talker (data transmission side) or a listener (data reception side) in accordance with the transmission schedule concerning the TAS traffic in the network. For example, in a combination of the end devices considering the RTA traffic of the STA **103** and the sensor apparatus **107**, the STA **103** operates as a listener and the sensor apparatus **107** operates as a talker. Details of the message sequence in this case will be described later with reference to FIG. 7, as an example of the flow of downlink (DL) communication of data communication supporting TAS in the BSS **101**. In a combination of the end devices considering the RTA traffic of the STA **103** and the remote apparatus **108**, the remote apparatus **108** operates as a listener and the STA **103** operates as a talker. Details of the message sequence in this case will be described later with reference to FIG. 8, as an example of the flow of uplink (UL) communication of data communication supporting TAS in the BSS **101**.

(27) Note that the above-described configuration of the network is merely an example, and the present invention is not limited to the configuration shown in FIG. 1. For example, the following discussion is applicable to, for example, a network (BSS) including many wireless communication apparatuses in a wider region, or the positional relationship between various wireless communication apparatuses. In addition, for example, the schedule management apparatus **106** sets a transmission/reception schedule. However, the function of the schedule management apparatus **106** may be included in a network node such as the AP **102**, or may be included in the STA **103** in some cases. That is, the AP **102** and the STA **103** may determine the transmission/reception schedule of data frames.

(28) As will be described later with reference to FIGS. 15 and 16, the apparatuses supporting the TAS scheme can transmit/receive TAS traffic data at a strict timing. Therefore, by setting an appropriate communication timing, it is possible to perform communication in a low-latency environment that is reliable, in some cases, such that there is no interference from another apparatus. On the other hand, it is also assumed that a communication apparatus supporting no TAS scheme exists in the area of the network, as a matter of course. Even if the schedule management apparatus **106** transmits a TAS scheme control message to the communication apparatus supporting no TAS scheme, the communication apparatus cannot interpret the message. This may cause the inconvenience of, for example, performing an unexpected operation.

(29) This embodiment provides, on that assumption, a method of making a notification of capability information of each communication apparatus to make it possible to confirm whether each communication apparatus supports the TAS scheme. More specifically, a notification of information is made by adding an information element to a MAC (Media Access Control) header of a radio frame to be transmitted by each communication apparatus. By selecting, based on the pieces of information, whether to use scheduling by the TAS scheme, it is possible to perform appropriate

scheduling between a transmission apparatus and a reception apparatus. The arrangements of the apparatuses that transmit/receive such notification, the procedure of processing, and the practical structure of the frame will be exemplified below.

(30) (Apparatus Arrangement)

(31) FIG. 2 shows the hardware arrangement of the communication apparatus (the AP, the STA, and other end devices) according to this embodiment. The communication apparatus includes, as an example of the hardware arrangement, a storage unit **201**, a control unit **202**, a function unit **203**, an input unit **204**, an output unit **205**, a communication unit **206**, and an antenna **207**.

(32) The storage unit **201** is formed by both or either of a ROM (Read Only Memory) and a RAM (Random Access Memory). The storage unit **201** stores, for example, programs for performing various operations (to be described later) and various kinds of information such as communication parameters for wireless communication. Note that a storage medium such as a flexible disk, a hard disk, an optical disk, a magneto-optical disk, a CD-ROM, a CD-R, a magnetic tape, a nonvolatile memory card, or a DVD may be used as the storage unit **201**, other than memories such as a ROM and a RAM.

(33) The control unit **202** is formed by one or more processors, for example, a processor such as a CPU or an MPU, an ASIC (Application Specific Integrated Circuit), a DSP (Digital Signal Processor), an FPGA (Field Programmable Gate Array), and the like. The CPU is an acronym for Central Processing Unit, and the MPU is an acronym for Micro Processing Unit. The control unit **202** controls the overall communication apparatus by executing the programs stored in the storage unit **201**. Note that the control unit **202** may control the overall communication apparatus in cooperation with the programs stored in the storage unit **201** and an OS (Operation System).

(34) The control unit **202** controls the function unit **203** to execute predetermined processing such as image capturing, printing, or projection. The function unit **203** is hardware used by the communication apparatus to execute predetermined processing. For example, in a case in which the communication apparatus is a camera, the function unit **203** is an image capturing unit and performs image capturing processing. For example, in a case in which the communication apparatus is a printer, the function unit **203** is a printing unit and performs print processing. For example, in a case in which the communication apparatus is a projector, the function unit **203** is a projection unit and performs projection processing. Data to be processed by the function unit **203** may be data stored in the storage unit **201**, or may be data communicated with another communication apparatus via the communication unit **206** (to be described later).

(35) The input unit **204** accepts various operations from a user. The output unit **205** performs various kinds of outputs to the user. In this example, the output by the output unit **205** includes at least one of display on a screen, audio output by a loudspeaker, vibration output, and the like. Note that both the input unit **204** and the output unit **205** may be implemented by one module, like a touch panel. The communication unit **206** controls wireless communication complying with the IEEE802.11 standard series, or controls IP communication. In the embodiment, the communication unit **206** can execute processing complying with at least the IEEE802.11be standard. In addition, the communication unit **206** controls the antenna **207** to transmit/receive radio signals for wireless communication. The communication apparatus communicates a content such as image data, document data, or video data with another communication apparatus via the communication unit **206**. The antenna **207** is an antenna capable of receiving signals in one of the sub-GHz band, 2.4-GHz band, 5-GHz band, and 6-GHz band. The radio antenna **207** may be constituted physically by one or more antennas in order to implement MIMO (Multi-Input and Multi-Output) transmission/reception.

(36) FIG. 3 shows an example of a functional arrangement associated with TAS communication of the communication apparatus. The communication apparatus includes, for example, a TAS capability information generation unit **301**, a TAS scheme determination unit **302**, a connection processing unit **303**, a MAC frame generation unit **304**, and a data transmission/reception unit **305**.

Note that these functional units can be implemented when, for example, the control unit **202** executes the programs stored in the storage unit **201**. However, the present invention is not limited to this and, for example, dedicated hardware components corresponding to the functional blocks shown in FIG. 3 may be used.

(37) The TAS capability information generation unit **301** generates information (TAS capability information) of the TAS capability of the communication apparatus, and stores it in the storage unit **201**. Based on the information concerning TAS received from the schedule management apparatus **106**, the TAS capability information generation unit **301** can reconstruct the TAS capability information stored in the storage unit **201** or update contents of the TAS capability information. The TAS capability information includes a support ID indicating the TAS scheduling scheme of the communication apparatus. The support ID is an identifier corresponding to each of a plurality of schemes, and is used to indicate which of the schemes can be used. The TAS capability information will be described later with reference to FIGS. 9 to 12. Note that the communication apparatus executes at least one of a notification of the capability information of the self-apparatus to another communication apparatus and reception of the capability information of the other communication apparatus from that apparatus. That is, the communication apparatus may only receive the capability information of another communication apparatus and need not notify the other communication apparatus of the capability information of the self-apparatus.

(38) The TAS scheme determination unit **302** determines, based on the TAS capability information of the communication partner apparatus, the TAS scheme to be used for communication with the partner apparatus. As the TAS schemes, there are generally two kinds of schemes of “shaping” and “policing”. Since the characteristics of these schemes are different from each other, the scheme corresponding to the use application (application) is used based on the difference between the characteristics. The “policing” scheme is a scheme of reducing latency caused by queueing by dropping a packet exceeding a limit rate or changing the priority of a packet. The “shaping” scheme buffers a packet exceeding a limit rate in an I/F queue, thereby causing latency. For example, in a case of traffic using a TCP/UDP port number, a scheme selection can be made to use the “shaping” scheme for TCP and to use the “policing” scheme for UDP. For example, if the TAS scheme usable by the STA is only “shaping”, “shaping” can be adopted, and if the TAS scheme usable by the STA is only “policing”, “policing” can be adopted. If the STA can use both the TAS schemes of “shaping” and “policing”, one of the TAS schemes is determined in accordance with the application feature of data transmitted/received to/from the STA. For example, the shaping scheme can be selected to transmit/receive data, such as sensor data, which is periodic but which is not restricted on latency or is not strictly restricted on latency. Alternatively, if an application which is restricted on a data arrival time because it requires a real time operation like a game or an industrial robot is used, the policing scheme can be selected so as to transmit/receive periodic data in consideration of latency control. Note that the shaping scheme and the policing scheme are merely examples, and another scheme may be usable. In this case, the TAS scheme determination unit **302** can select a scheme to be used from a plurality of usable schemes including the other scheme.

(39) The TAS capability information can include information of the TAS scheme corresponding to the application feature of data to be transmitted/received, and the communication apparatus notifies the partner apparatus of the TAS capability information using, for example, a management frame defined by the IEEE802.11 standard at the time of connection or during communication. The information notification by the management frame will be described later with reference to FIG. 7. If the number of users (end devices) accommodated in one BSS is large, overhead may become large due to exchange of information for low-latency communication. To cope with this, the kind of the TAS scheme may be changed when the number of STAs accommodated by the AP is equal to or larger than a predetermined number. Note that the communication apparatus may or may not notify the communication partner apparatus of the TAS scheme determined by the TAS scheme

determination unit **302**.

(40) The connection processing unit **303** performs processing for establishing connection between the STA and the AP. For example, the connection processing unit **303** of the STA transmits an Association Request frame to the AP. The connection processing unit **303** of the AP transmits an Association Response frame as a response to the Association Request frame. The MAC frame generation unit **304** generates a MAC frame that stores the TAS capability information generated by the TAS capability information generation unit **301**, as needed. The MAC frame in this example is, for example, a MAC frame in a radio frame such as a Beacon frame, a Probe Request/Response frame, or an Association Request/Response frame. Alternatively, the MAC frame may be a MAC frame of a Reassociation Request/Response frame. Note that the TAS capability information can be stored in, for example, a portion of a MAC header. The TAS capability information or the TAS scheme information is transmitted using a capability element to be described later with reference to FIG. **9** or **11**. The data transmission/reception unit **305** transmits/receives a TAS traffic data frame generated periodically at a strict timing in accordance with the TAS scheme determined by the TAS scheme determination unit **302**. The TAS traffic data transmission processing in the data transmission/reception unit **305** will be described in detail later with reference to FIGS. **15** and **16**.

(41) (Procedure of Processing)

(42) An example of the procedure of processing executed by the AP **102** when the STA **103** is connected to the AP **102** will be described with reference to FIG. **4**. Note that the same processing can be executed even for another combination of the STA and the AP. This processing can be started when, for example, the STA **103** is powered on. Alternatively, this processing may be started when the user or application instructs, in the STA **103**, to start data communication in which the TAS scheme should be used. Note that this processing is implemented when, for example, the control unit **202** of the AP **102** executes the program stored in the storage unit **201**.

(43) In this processing, the AP **102** obtains the TAS capability information of the STA **103** (step **S401**). For example, the TAS capability information shown in FIG. **9** or **10** (to be described later) is included in a Probe Request frame to be transmitted before the STA **103** is connected or an Association Request frame to be transmitted at the time of connection. The AP **102** can obtain the TAS capability information of the STA **103** by receiving the frame and analyzing its contents. Subsequently, the TAS scheme determination unit **302** of the AP **102** determines, based on the TAS capability information of the STA **103** obtained in step **S401**, the TAS scheme to be used for latency control (step **S402**). Then, the AP **102** determines whether the TAS scheme determined in step **S402** can be used (step **S403**). If it is determined that the TAS scheme determined in step **S402** cannot be used (NO in step **S403**), the AP **102** notifies the STA **103** that latency control cannot be executed. In this case, the AP **102** notifies the STA **103** of, for example, a management frame including the TAS capability information in which information (for example, an ID) indicating that latency control cannot be executed is set (step **S406**). On the other hand, if it is determined that the TAS scheme determined in step **S402** can be used (YES in step **S403**), the TAS capability information generation unit **301** of the AP **102** notifies the STA **103** of the TAS scheme determined in step **S402**. In this case, the AP **102** notifies the STA **103** of, for example, a management frame including the TAS capability information in which a value indicating the TAS scheme determined in step **S402** is set (step **S404**). Note that this value can be, for example, the TAS support ID corresponding to the TAS scheme. Then, the AP **102** stores, in the storage unit **201**, the TAS capability information included in the management frame and notified to the STA **103** in step **S404** or **S406** (step **S405**). In this way, the TAS capability information is exchanged between the AP and the STA.

(44) Subsequently, an example of the procedure of data reception processing executed by the STA **103** at the time of communication in a downlink (DL) as a link used to transmit a signal from the AP **102** to the STA **103** will be described with reference to FIG. **5**. In this example, processing of TAS traffic data (sensor data or the like) when the apparatus such as the sensor apparatus **107**

serves as a talker (transmission side) and the STA **103** serves as a listener (reception side) will be explained. For example, TAS traffic data periodically transmitted from the sensor apparatus **107** reaches the STA **103** via the bridges **104** and the AP **102**. Note that this processing is also applicable to an uplink (UL) as a link used to transmit a signal from the STA **103** to the AP **102**. The same processing can also be executed when the AP **102** receives data from the STA **103**. In this case, in the following description, the STA **103** can be replaced by the AP **102**.

(45) When receiving data from the AP **102** (YES in step **S501**), the STA **103** determines whether the self-apparatus operates in a TAS reception mode such as policing or shaping. The processing of determining whether the self-apparatus operates in the TAS reception mode can be performed by, for example, confirming the TAS support ID stored in the storage unit **201** when the STA **103** receives the management frame from the AP **102**. If the self-apparatus does not operate in the TAS reception mode such as policing or shaping (NO in step **S502**), the STA **103** executes normal data reception processing (step **S503**). On the other hand, if the self-apparatus operates in the TAS reception mode (YES in step **S502**), the STA **103** sets a TAS traffic data reception timer with a time interval shorter than the next TAS traffic data reception timing after the data reception processing, and activates the timer (step **S504**).

(46) To determine whether the current time is the TAS traffic data reception timing, the STA **103** checks the TAS traffic data reception timer (step **S505**). If the current time is not the next TAS traffic data reception timing (NO in step **S505**), the STA **103** checks, for example, whether a predetermined time such as a timing immediately before reception has been reached (step **S508**). If the predetermined time has not been reached (NO in step **S508**), the STA **103** returns the process to step **S505**, and checks the TAS traffic data reception timing. On the other hand, if the predetermined time has been reached (YES in step **S508**), the STA **103** executes a carrier sense or the like to confirm the use state of a corresponding TAS traffic data reception channel (step **S509**). If it is determined that the reception channel is in a busy state (NO in step **S509**), the STA **103** executes channel access contention processing (step **S510**) to set the channel in an available state, and returns the process to step **S505**. If the reception channel is not busy but available (YES in step **S509**), the current time is immediately before the reception timing, and thus the STA **103** reserves the TAS traffic data reception channel by, for example, transmitting a null packet (step **S511**). Then, the STA **103** returns the process to step **S505**.

(47) If the STA **103** checks the TAS traffic data reception timer, and determines that the current time is the TAS traffic data reception timing (YES in step **S505**), the STA **103** executes TAS traffic data reception processing (step **S506**). Then, the STA **103** determines whether data communication ends (step **S507**). If data communication continues (NO in step **S507**), the STA **103** returns the process to step **S501**. If data communication ends (YES in step **S507**), the STA **103** ends the data reception processing. As described above, the STA **103** can receive the TAS traffic data from the AP **102** at the predetermined reception timing (time) in wireless communication of the BSS **101**.

(48) An example of the procedure of data transmission processing executed by the STA **103** at the time of communication in the uplink (UL) will be described with reference to FIG. **6**. In this example, processing of TAS traffic data (remote control data or the like) when the STA **103** serves as a talker (transmission side) and the apparatus such as the remote apparatus **108** serves as a listener (reception side) will be explained. For example, TAS traffic data periodically transmitted from the STA **103** reaches the remote apparatus **108** via the AP **102** and the bridges **104**. Note that this processing is also applicable to the downlink (DL). The same processing can also be executed when, for example, the AP **102** transmits data from the STA **103**. In this case, in the following description, the STA **103** can be replaced by the AP **102**.

(49) The STA **103** continuously monitors whether transmission data is generated by, for example, receiving a data transmission request from the control application of the remote apparatus **108** implemented in the self-apparatus (step **S601**). Then, if generation of transmission data is detected (YES in step **S601**), the STA **103** determines whether the self-apparatus operates in a TAS

transmission mode such as policing or shaping. The processing of determining whether the self-apparatus operates in the TAS transmission mode can be performed by, for example, confirming the TAS support ID stored in the storage unit **201** when the STA **103** receives the management frame from the AP **102**. If the self-apparatus does not operate in the TAS transmission mode such as policing or shaping (NO in step **S602**), the STA **103** executes normal data transmission processing (step **S603**). On the other hand, if the self-apparatus operates in the TAS transmission mode (YES in step **S602**), the STA **103** sets a TAS traffic data transmission timer with a time interval shorter than the next TAS traffic data transmission timing after the data transmission processing, and activates the timer (step **S604**).

(50) To determine whether the current time is the TAS traffic data transmission timing, the STA **103** checks the TAS traffic data transmission timer (step **S605**). If the current time is not the next TAS traffic data transmission timing (NO in step **S605**), the STA **103** checks, for example, whether a predetermined time such as a timing immediately before transmission has been reached (step **S608**). If the predetermined time has not been reached (NO in step **S608**), the STA **103** returns the process to step **S605**, and checks the TAS traffic data transmission timing. On the other hand, if the predetermined time has been reached (YES in step **S608**), the STA **103** executes a carrier sense or the like to confirm the use state of a corresponding TAS traffic data transmission channel (step **S609**). If it is determined that the transmission channel is in a busy state (NO in step **S609**), the STA **103** executes channel access contention processing (step **S610**) to set the channel in an available state, and returns the process to step **S605**. If the transmission channel is not busy but available (YES in step **S609**), the current time is immediately before the transmission timing, and thus the STA **103** reserves the TAS traffic data transmission channel by, for example, transmitting a null packet (step **S611**). Then, the STA **103** returns the process to step **S605**.

(51) If the STA **103** checks the TAS traffic data transmission timer, and determines that the current time is the TAS traffic data transmission timing (YES in step **S605**), the STA **103** executes TAS traffic data transmission processing (step **S606**). The TAS traffic data transmission processing will be described later with reference to FIGS. **15** and **16**. After that, the STA **103** determines whether data communication ends (step **S607**). If data communication continues (NO in step **S607**), the STA **103** returns the process to step **S601**. If data communication ends (YES in step **S607**), the STA **103** ends the data transmission processing. As described above, the STA **103** can transmit the TAS traffic data to the AP **102** at the predetermined transmission timing (time) in wireless communication of the BSS **101**.

(52) As described above, the STA **103** can transmit the TAS traffic data to the AP **102** at the predetermined transmission timing. By executing the processes shown in FIGS. **5** and **6** between the STA and the AP, the TAS traffic data is periodically transmitted at the preset time. When an operation is executed so that communication is performed at the time preset appropriately based on the network topology, an unnecessary standby time is reduced, thereby making it possible to implement low-latency communication.

(53) Subsequently, an example of the flow of messages transmitted/received between the AP **102** and the STA **103** at the time of downlink (DL) communication will be described with reference to FIG. **7**. This processing is executed when TAS traffic data from an apparatus such as the sensor apparatus **107** operating as a talker is received by the STA **103** as a listener via the bridges **104** and the AP **102**. First, the STA **103** executes scan processing for obtaining network information of the AP **102**. For example, the AP **102** broadcasts a Beacon frame (M701) including the network information to apparatuses in the area of the BSS **101**. At this time, the AP **102** may broadcast a Beacon frame including TAS scheme capability information to be described later with reference to FIG. **9** or **10**. The STA **103** may transmit a Probe Request frame (M702) for inquiring about the network information of the AP **102**, and obtains the information from the AP **102**. In response to the Probe Request frame (M702), the AP **102** transmits a Probe Response frame (M703). For example, the STA **103** can receive the Beacon frame (M701) transmitted by the AP **102**, and obtain

the network information of the AP **102** from the Beacon frame. Alternatively, the STA **103** may actively transmit the Probe Request frame (M702), and receive the Probe Response frame (M703) from the AP **102**, thereby obtaining the network information of the AP **102**. At this time, the Beacon frame (M701) and the Probe Response frame (M703) include, as TAS capability information, information indicating whether the AP **102** supports the TAS scheme. If the TAS scheme is supported, the TAS capability information included in the frame can indicate whether only the policing scheme is supported, only the shaping scheme is supported, or both the schemes are supported. The STA **103** can include the TAS capability information of the STA **103** in the Probe Request frame (M702) and transmit it. With these processes, the STA **103** and the AP **102** can exchange the TAS capability information. However, the TAS capability information may be exchanged by another message such as an Association Request/Response frame (to be described later) without exchanging the TAS capability information at this timing.

(54) After the scan processing, the STA **103** transmits an Association Request frame (M704) to the AP **102** to be connected to the BSS **101**. In response to the Association Request frame (M704), the AP **102** transmits, to the STA **103**, an Association Response frame (M705) indicating the connection result of the STA **103**. The association Request frame and the Association Response frame may include the TAS capability information. The STA **103** may determine the TAS capability information to be included in the Association Request frame based on the TAS capability information of the AP **102** obtained by the scan processing. In one example, when both the shaping scheme and the policing scheme can be used, the STA **103** may select, in accordance with the application of communication by the TAS scheme, a scheme to be used, and notify the AP **102** of information indicating that only the selected scheme can be used. If, for example, the sensor apparatus **107** serves as a talker, the STA **103** that can use both the schemes may transmit, to the AP **102**, the Association Request frame (M704) including the TAS capability information indicating that the self-apparatus can use only the shaping scheme. Alternatively, if the remote apparatus **108** serves as a listener, the STA **103** that can use both the schemes may transmit, to the AP **102**, the Association Request frame (M704) including the TAS capability information indicating that the self-apparatus can use only the policing scheme. Similarly, the AP **102** may determine the TAS capability information to be included in the Association Response frame based on the TAS capability information of the STA **103** included in the Association Request frame. In this way, the TAS capability information is exchanged before connection between the AP **102** and the STA **103** is established.

(55) After that, after confirming that TAS communication is possible, the AP **102** transmits a management frame (M706) to the STA **103**. In this management frame, for example, information indicating that the shaping scheme is to be used is included in the TAS capability information at the time of DL data communication. This management frame can include, for example, information indicating a schedule according to which data are to be transmitted/received by the TAS scheme. After that, the STA **103** serving as a listener periodically receives, via the AP **102**, TAS traffic data (M707) from the sensor apparatus **107** serving as a talker. For example, the STA **103** executes the data reception processing described with reference to FIG. 5, thereby receiving the TAS traffic data (M707). For example, the AP **102** executes the data transmission processing described with reference to FIG. 6, thereby transmitting the TAS traffic data (M707).

(56) Subsequently, an example of the flow of messages transmitted/received between the AP **102** and the STA **103** at the time of UL communication will be described with reference to FIG. 8. This processing is executed when the STA **103** serving as a talker transmits TAS traffic data that is received by an apparatus such as the remote apparatus **108** via the AP **102** and the bridges **104**. In this example, the messages M701 to M703 concerning scan processing, the messages M704 to M705 concerning STA connection processing, and the management frame M706 are the same as in FIG. 7. Note that since the remote apparatus **108** serves as a listener, the STA **103** that can support both the schemes can include, in the Association Request frame, the TAS capability information

indicating that the self-apparatus can use only the policing scheme, and transmit the Association Request frame to the AP **102**. Note that in the management frame (**M706**), for example, information indicating that the policing scheme is to be used is included in the TAS capability information at the time of UL data communication. This management frame can include, for example, information indicating a schedule according to which data are to be transmitted/received by the TAS scheme. After that, the STA **103** serving as a talker periodically transmits, via the AP **102**, TAS traffic data (**M801**) to the remote apparatus **108** serving as a listener. For example, the STA **103** executes the data transmission processing described with reference to FIG. **6**, thereby transmitting the TAS traffic data (**M801**). For example, the AP **102** executes the data reception processing described with reference to FIG. **5**, thereby receiving the TAS traffic data (**M801**).

(57) As described above, the AP **102** and the STA **103** can exchange the TAS capability information of the self-apparatus, and can determine whether communication by the TAS scheme is performed in communication thereafter, and which of the shaping scheme and the policing scheme is to be used. Therefore, by executing the processing to be described later with reference to FIGS. **15** and **16**, it is possible to perform wireless communication by scheduling according to the time, thereby implementing low-latency communication.

(58) Next, the first example of the TAS capability information will be described with reference to FIG. **9**. Note that in this embodiment, the name of the TAS capability information is “TAS capability element” but the present invention is not limited to this. For example, another name such as “TAS Element” may be used. The TAS capability element has the same structure as that of another Information Element defined by the IEEE802.11 standard. That is, the TAS capability element is formed by including an Element ID field **901** for identifying an Element, a Length field **902** indicating the data length of the Element, and Element-specific information. The TAS capability element includes a TAS capability Info field **903** as the Element-specific information. The TAS capability element is included in, for example, a MAC frame such as a Beacon frame, a Probe Request/Response frame, or an Association Request/Response frame. Note that the TAS capability element may be included in a MAC frame of a Reassociation Request/Response frame. The TAS capability Info field **903** includes information indicating whether the transmitter of the TAS capability element can use the TAS scheme, and the usable TAS scheme. This information is represented by, for example, 2 bits. An example of the information represented by 2 bits will be described later with reference to FIG. **11**. Note that the TAS capability Info field **903** has a size of 1 octet (8 bits) in this example but the present invention is not limited to this. The names of the fields and the positions and sizes of bits are not limited to the example shown in FIG. **9** and pieces of similar information may be stored with different field names in a different order and different sizes.

(59) The second example of the TAS capability information will be described with reference to FIG. **10**. Similar to the example shown in FIG. **9**, the TAS capability element includes the Element ID field **901** and the Length field **902**. This information can be included in a MAC frame of each radio frame described above, instead of the information shown in FIG. **9**. The TAS capability element shown in FIG. **10** can be the same as that shown in FIG. **9** except for a TAS capability Info field **1001**. However, since the size of the TAS capability Info field is different, a value stored in the Length field **902** is different between FIGS. **9** and **10**. In the example shown in FIG. **10**, the TAS capability Info field **1001** is formed by information indicating availability of the TAS scheme in each primary channel in addition to availability of the TAS scheme or the like. The primary channel is a main channel for supporting a multi-link technique of using a plurality of radio channels at the same time, and is used to control transmission/reception of capability information of another radio link and messages associated with connection and disconnection. Note that the relationship between the primary channel and the TAS scheme capability information will be described later with reference to FIG. **14**. Note that the primary channel will sometimes be referred to as “PCH” hereinafter.

(60) A TAS support field **1002** includes information indicating whether the transmitter of the TAS

capability element can use the TAS scheme, and the usable TAS scheme. In one example, this information is represented by 2 bits. An example of the information represented by 2 bits will be described later with reference to FIG. 11. A PCH 1 availability field **1003**, a PCH 2 availability field **1004**, a PCH 3 availability field **1005**, a PCH 4 availability field **1006**, a PCH 5 availability field **1007**, a PCH 6 availability field **1008**, a PCH 7 availability field **1009**, a PCH 8 availability field **1010**, and a PCH 9 availability field **1011** are fields each indicating availability of the TAS scheme in each of primary channels 1 to 9. A value stored in each of these fields and contents indicated by the value will be described later with reference to FIG. 12. A Reserve field **1012** is a 5-bit unused region for future extension. An example in which the TAS capability Info field **1001** has a size of 2 octets (16 bits) is shown but the present invention is not limited to this. For example, pieces of similar information may be indicated by different field names at different bit positions in different field sizes. Note that in this embodiment, the name of the element shown in FIG. 10 is the TAS capability element. However, the present invention is not limited to this, and for example, another name such as a TAS Multi-Link Element may be used.

(61) An example of the information indicating whether the transmitter of the information can use the TAS scheme, and the usable TAS scheme in the TAS capability information shown in FIG. 9 or 10 will be described with reference to FIG. 11. Note that the information indicating whether the TAS scheme can be used and the usable TAS scheme will sometime be referred to as a TAS support status hereinafter. Referring to FIG. 11, a D-bit value **1101** represents a 2-bit data bit (D-bit) value stored in the TAS capability Info field **903** shown in FIG. 9 or the TAS support field **1002** shown in FIG. 10. Note that a bit string represented by the D-bit value can be a TAS support ID. TAS support contents **1102** indicate contents corresponding to each TAS support ID. For example, a bit value “00” of the D-bit value **1101** indicates that the TAS scheme is not supported (unavailable). A bit value “01” of the D-bit value **1101** indicates that the TAS scheme is supported but only the shaping scheme can be used. A bit value “10” of the D-bit value **1101** indicates that the TAS scheme is supported but only the policing scheme can be used. A bit value “11” of the D-bit value **1101** indicates that the TAS scheme is supported and both the policing scheme and the shaping scheme can be used. Note that the setting value concerning each information element is not limited to this example, and pieces of similar information may be indicated by different field names or different values.

(62) FIG. 12 shows an example of a value stored in each of the PCH 1 availability field **1003**, the PCH 2 availability field **1004**, the PCH 3 availability field **1005**, the PCH 4 availability field **1006**, the PCH 5 availability field **1007**, the PCH 6 availability field **1008**, the PCH 7 availability field **1009**, the PCH 8 availability field **1010**, and the PCH 9 availability field **1011** shown in FIG. 10 and contents corresponding to the value. In each of the PCH 1 availability field **1003**, the PCH 2 availability field **1004**, the PCH 3 availability field **1005**, the PCH 4 availability field **1006**, the PCH 5 availability field **1007**, the PCH 6 availability field **1008**, the PCH 7 availability field **1009**, the PCH 8 availability field **1010**, and the PCH 9 availability field **1011**, an E-bit **1201** which is formed so that contents **1202** are different depending on a combination with the value in the TAS support field **1002** is stored. For example, if the two bits stored in the TAS support field **1002** indicate the policing scheme (D-bit value “10”) or the shaping scheme (D-bit value “01”), the E-bit represents availability of each TAS scheme. For example, when “1” is set in the E-bit, it can be indicated that the TAS scheme represented by the D-bit value can be used. When “0” is set in the E-bit, it can be indicated that the TAS scheme represented by the D-bit value cannot be used. Note that this is merely an example, and it may be indicated that the TAS scheme represented by the D-bit value can be used by setting “0” in the E-bit, and that the TAS scheme represented by the D-bit value cannot be used by setting “1” in the E-bit. Furthermore, if the two bits stored in the TAS support field **1002** indicate that both the policing scheme and the shaping scheme can be used (if the D-bit value is “11”), one of the schemes can be designated by one bit of the E-bit. For example, the policing scheme is designated by setting “1” in the E-bit, and the shaping scheme is designated

by setting “0” in the E-bit. Alternatively, the policing scheme may be designated by setting “0” in the E-bit, and the shaping scheme may be designated by setting “1” in the E-bit.

(63) An example of the procedure of processing executed by the AP **102** when broadcasting the TAS capability information shown in FIG. **10** by a Beacon frame will be described with reference to FIG. **13**. This processing is started when, for example, the AP **102** is powered on or collection processing of data of the TAS scheme is activated in the AP **102**. Note that this processing can be implemented when, for example, the control unit **202** of the AP **102** executes the program stored in the storage unit **201**.

(64) The AP **102** determines whether it is set in a mode of including the multi-channel TAS information shown in FIG. **10** in a Beacon frame (step **S1301**). If it is determined that the mode of including the multi-channel TAS information in a Beacon frame is not set (NO in step **S1301**), the AP **102** executes conventional Beacon frame setting processing (step **S1302**). Then, the AP **102** broadcasts, in the BSS **101**, the Beacon frame obtained by the setting processing in step **S1302** (step **S1306**). Note that in this case, the AP **102** may transmit, for example, the Beacon frame including the TAS information shown in FIG. **9**.

(65) If the mode of including the multi-channel TAS information in the Beacon frame is set (YES in step **S1301**), the AP **102** determines availability of each of the plurality of PCHs based on the TAS scheme capability information stored in step **S405** of FIG. **4** (step **S1303**). To do this, the AP **102** transmits an RTS (Request To Send) message. Then, upon receiving the RTS message in each PCH, the STA **103** executes a carrier sense in the PCH. If the PCH is not in the busy state, the STA **103** transmits a CTS (Clear To Send) message to the AP **102** after a lapse of a SIFS (Short Inter Frame Space) period. The AP **102** confirms whether the CTS message is received in each PCH in which the RTS message has been transmitted, and recognizes, as an unavailable PCH, the PCH for which the reception of the CTS message is confirmed. At this time, with respect to the PCH for which the reception of the CTS message cannot be confirmed, the TAS capability information generation unit **301** of the AP **102** recognizes that TAS communication cannot be used.

(66) Then, the TAS capability information generation unit **301** updates the information of availability of the TAS scheme concerning each PCH in the TAS capability information. When the processing in step **S1303** is complete for all the plurality of available PCHs (YES in step **S1304**), the AP **102** stores the confirmed TAS capability information in the storage unit **201**, and sets the TAS capability information in the Beacon frame (step **S1305**). The AP **102** broadcasts, in the BSS **101**, the Beacon frame including the TAS capability information (step **S1306**). Note that the Beacon frame including the TAS capability information shown in FIG. **9** or **10** may be broadcasted in the plurality of PCHs.

(67) As described above, the AP **102** can include the information of the plurality of PCHs in the Beacon frame and transmit it. This allows, for example, the STA **103** to select a PCH in which communication can be performed in consideration of latency control by the TAS capability information and use it. Among the PCHs having the TAS capability information, a PCH in which the size of a signal bandwidth is equal to or larger than a predetermined width may be selected or a PCH in which the signal bandwidth is large may be preferentially selected. Among the PCHs in the TAS capability information, a PCH in which the reception field intensity or the signal-to-noise ratio (SNR) is higher than a predetermined value may be selected, or a PCH in which these values are satisfactory may be preferentially selected. Among the PCHs in the TAS capability information, a PCH in which a channel transmission capacity calculated from the bandwidth and SNR is equal to or larger than a predetermined value may be selected, or a PCH in which the capacity is large may be preferentially selected. Among the PCHs in the TAS capability information, a PCH to be used may be selected based on free channel information estimated from a channel use status in a predetermined period. A condition other than those PCH selection conditions may be used. Furthermore, one or more of the above-described selection conditions may be used in combination.

(68) PCHs 1 to 9 will be described with reference to FIG. **14**. PCHs 1 to 9 indicate a plurality of

PCHs available in the 5-GHz band. For example, PCHs 1 to 9 are frequency channels of a 20-MHz bandwidth with channel numbers of 36, 44, 52, 60, 100, 108, 116, 124, and 132. Note that the 5-GHz frequency band has been explained but the same discussion is also applicable to 2.4- or 6-GHz frequency band. The above-described discussion may be applied not only to the PCHs but also to another arbitrary frequency channel.

(69) Time-Aware Scheduling (TAS) traffic data transmission processing in the AP and the STA according to this embodiment will be described with reference to FIG. 15. As one example, the AP 102 and the STA 103 have an arrangement shown in FIG. 15 for the TAS traffic data transmission processing. In the arrangement shown in FIG. 15, an application 1501 receives TAS scheme information determined by the above-described TAS scheme determination unit 302 before a state in which communication of TAS traffic data is in progress is set. To execute the transmission processing in accordance with the TAS scheme information, the application 1501 prepares for a data transmission request from the MAC frame generation unit 304. A traffic classification function 1503 receives traffic data added with priority information from the MAC frame generation unit 304, and maps temporally restricted traffic data in a TAS queue (Q0) 1504. The traffic classification function 1503 maps best effort traffic data other than the TAS traffic data in each of BE queues (Q1 to Q4) 1505. A TAS time gate control function 1502 temporarily stops/resumes each of a TAS time gate (G0) 1506 and various BE time gates (G1 to G4) 1507, thereby executing output control of the traffic data accumulated in each queue. For example, as will be described later, the TAS time gate control function 1502 executes control to protect the TAS traffic data while avoiding channel access contention.

(70) TAS traffic data transmission processing using the plurality of time gates (G0 to G4) will be described with reference to FIG. 16. A curve 1601 indicates the OPEN/CLOSE state of the TAS time gate (G0) 1506, and curves 1602 to 1605 indicate the OPEN/CLOSE states of the various BE time gates (G1 to G4) 1507, respectively. In these curves, “1” indicates the OPEN state, and “0” indicates the CLOSE state. If the time gate is opened, the data in the queue corresponding to the time gate can be output via the time gate. A guard time 1606 indicates a time T_g during which transmission of all the gates is blocked. This guard time 1606 can prevent various BE traffic data from colliding against the TAS traffic data. A timing 1607 indicates a timing T_o at which the TAS time gate (G0) 1506 is periodically opened. The timing T_o is generated for every cycle 1608. The cycle 1608 is defined by an interval T_p at which data is transmitted at a strict and periodic timing designated by a TAS traffic data transmission request. A period 1609 indicates a time section T_c during which the TAS time gate (G0) 1506 of the TAS queue (Q0) 1504 is opened, and indicates a period during which the TAS traffic data is transmitted at a strict and periodic timing.

(71) The TAS time gate control function 1502 executes gate control corresponding to the transmission/reception schedule of the TAS traffic data which is periodically repeated. The TAS time gate control function 1502 can close or open each time gate by, for example, inputting a bit of 0 or 1 to each time gate. The TAS time gate control function 1502 can indicate the OPEN/CLOSE states of the five time gates G0 to G4 by a 5-bit string. Then, each time gate can extract a bit corresponding to itself from the bit string, and execute control of the CLOSE/OPEN state. The TAS time gate control function 1502 may output, to each time gate, 1-bit information indicating whether the time gate is closed or opened. The states of the time gates for each bit string will now be described. Note that the bit string indicates the OPEN/CLOSE states of the time gates in the order of “G0.Math.G1.Math.G2.Math.G3.Math.G4”. A bit string “00000” indicates a data block state. This state corresponds to the guard time 1606, and all the time gates (G0 to G4) are closed. A bit string “10000” indicates a state in which only the TAS time gate (G0) 1506 is opened, and this state corresponds to the period 1609. A bit string “01111” indicates a state in which the various BE time gates (G1 to G4) 1507 are opened, and the state corresponds to, for example, a period obtained by excluding the period 1609 and the guard time 1606 from the cycle 1608. In this way, the TAS time gate control function 1502 can control the CLOSE/OPEN state of each time gate by a 1-bit

indicator.

(72) As described above, at the transmission timing of TAS traffic data which should be transmitted at a strict timing, the TAS traffic data can be processed without latency using the time gates (G0 to G4) while blocking data which should not be transmitted.

(73) According to the present invention, it is possible to implement reliable low-latency communication.

Other Embodiments

(74) Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

(75) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

Claims

1. A communication apparatus operating as an access point, comprising: at least one memory that stores a set of instructions; and at least one processing circuit, wherein the communication apparatus is caused, by the at least one processing circuit executing the instructions and/or the at least one processing circuit itself operating, to perform operations comprising: transmitting a radio frame complying with an IEEE802.11 standard to another communication apparatus, wherein a MAC (Media Access Control) frame of the radio frame transmitted includes an information element indicating whether the communication apparatus supports a scheme of scheduling of transmission opportunities in which contention is restricted based on a time to perform a low-latency communication, wherein the radio frame including the MAC frame including the information element indicating whether the communication apparatus supports the scheme of scheduling of the transmission opportunities is one of a Beacon frame a Probe Response frame, an Association Response frame, and a Reassociation Response frame.

2. The communication apparatus according to claim 1, wherein the information concerning the scheduling of transmission opportunities in which contention is restricted based on the time includes information indicating which of a plurality of schemes of the scheduling of transmission opportunities in which contention is restricted based on the time can be used by the communication apparatus that has transmitted the radio frame.

3. The communication apparatus according to claim 1, wherein the MAC frame includes an

element containing one ID subfield and multiple fields including the information concerning the scheduling of transmission opportunities in which contention is restricted based on the time for each of a plurality of frequency channels.

4. The communication apparatus according to claim 3, wherein the MAC frame includes a first value indicating which of a plurality of schemes of the scheduling of transmission opportunities in which contention is restricted based on the time can be used by the communication apparatus that has transmitted the radio frame, and a second value for each of the plurality of frequency channels, and a combination of the first value and the second value indicates whether the scheduling of transmission opportunities in which contention is restricted based on the time is available or not, and indicates a scheduling scheme in a case where the scheduling of transmission opportunities in which contention based on the time is restricted is available.

5. The communication apparatus according to claim 1, wherein when transmitting the radio frame including the information concerning the scheduling of transmission opportunities in which contention is restricted based on the time, based on an application that performs communication using transmission opportunities in which contention is restricted based on the time, the communication apparatus sets the information concerning the scheduling of low latency data communication based on the time.

6. The communication apparatus according to claim 1, wherein the information concerning the scheduling of transmission opportunities in which contention is restricted based on the time is transmitted to the other communication apparatus or received from the other communication apparatus before connection to the other communication apparatus is established.

7. The communication apparatus according to claim 1, wherein the information concerning the scheduling of transmission opportunities in which contention is restricted based on the time indicates scheduling for transmitting or receiving a data frame to or from the other communication apparatus.

8. The communication apparatus according to claim 7, wherein the information concerning the scheduling of transmission opportunities in which contention is restricted based on the time includes a scheme to be used to transmit or receive the data frame, which is set based on a scheme of the scheduling of transmission opportunities in which contention is restricted based on the time that can be used by the other communication apparatus and a scheme of the scheduling of transmission opportunities in which contention is restricted based on the time that can be used by the communication apparatus.

9. The communication apparatus according to claim 1, wherein if the communication apparatus cannot use a scheme of the scheduling of transmission opportunities in which contention is restricted based on the time that can be used by the other communication apparatus, the information concerning the scheduling based on the time includes information indicating that the scheduling of transmission opportunities in which contention is restricted based on the time cannot be used.

10. The communication apparatus according to claim 1, wherein the other communication apparatus is an access point apparatus, and after the radio frame including information relating to the scheduling of transmission opportunities in which contention is restricted based on the time is transmitted to the access point apparatus, in a case where a schedule determined by the access point apparatus is received from the access point apparatus, the communication apparatus transmits a data frame based on the received schedule.

11. A control method executed by a communication apparatus operating as an access point, comprising: transmitting a radio frame complying with an IEEE802.11 be standard to another communication apparatus, wherein a MAC (Media Access Control) frame of the radio frame transmitted includes an information element indicating whether the communication apparatus supports a scheme of scheduling of transmission opportunities in which contention is restricted based on a time to perform a low-latency communication, wherein the radio frame including the MAC frame including the information element indicating whether the communication apparatus

supports the scheme of scheduling of the transmission opportunities is one of a Beacon frame a Probe Response frame, an Association Response frame, and a Reassociation Response frame.

12. A non-transitory computer-readable storage medium that stores a program for causing a computer included in a communication apparatus to: transmit a radio frame complying with an IEEE802.11 be standard to another communication apparatus, wherein a MAC (Media Access Control) frame of the radio frame transmitted includes an information element indicating whether the communication apparatus supports a scheme of scheduling of transmission opportunities in which contention is restricted based on a time to perform a low-latency communication, wherein the radio frame including the MAC frame including the information element indicating whether the communication apparatus supports the scheme of scheduling of the transmission opportunities is one of a Beacon frame a Probe Response frame, an Association Response frame, and a Reassociation Response frame.
